

# Software Engineering Lab 2 — Agile Backlog & Sprint Report

## Digital Gift Card & Loyalty Point Exchange

### Problem Statement #34 — Digital Gift Card & Loyalty Point Exchange

|                  |                                                    |
|------------------|----------------------------------------------------|
| <b>Student</b>   | Girish G N                                         |
| <b>SRN</b>       | PES1UG24CS571                                      |
| <b>Tool Used</b> | Jira (Software Project)                            |
| <b>Lab</b>       | Lab 2 — Agile Backlog Creation & Sprint Simulation |
| <b>Duration</b>  | 60 minutes (per handout)                           |

*This report documents the Agile transformation of the Lab 1 functional requirements into a Jira backlog, including Epics, User Stories, Story Point estimation, a two-sprint plan, sprint execution, and burndown analysis for Lab 2.*

## Introduction & Scope

This report presents the Lab 2 deliverables for the project **Digital Gift Card & Loyalty Point Exchange** (Problem Statement #34). The objective of Lab 2, as stated in the handout, was to use Jira to convert the Lab 1 functional requirements into an Agile backlog, simulate a sprint, and analyse progress using Epics, User Stories, prioritization, Story Points, the Planning Poker concept, a sprint workflow (To Do → In Progress → Done), and a Burndown Chart.

The report is organized into six parts, clearly separating information taken directly from the Lab 1 requirements document from the Agile decisions and estimates made during Lab 2:

- **Part A** — Lab 1 source requirements (Functional and Non-Functional)
- **Part B** — Agile backlog interpretation (Epics and User Stories)
- **Part C** — Story Point estimates (Fibonacci scale)
- **Part D** — Sprint plan (Sprint 1 and Sprint 2)
- **Part E** — Burndown chart and sprint execution evidence
- **Part F** — Reflection

**Note on scope:** All content in Part A is reproduced exactly as given in the Lab 1 requirements. All content in Parts B–E represents Agile artifacts and estimates created for the purposes of the Lab 2 simulation and is not part of the original Lab 1 requirements document.

## PART A

## Part A — Lab 1 Source Requirements

The following functional and non-functional requirements were defined in Lab 1 and form the factual basis for the Agile backlog created in Lab 2. Values below are reproduced exactly as given in the Lab 1 requirements table.

### A.1 Functional Requirements

| FR-001                     | Functional                                                                                                                                        | HIGH |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b>         | The system shall validate merchant loyalty points, convert them into universal exchange tokens, and generate single-use digital voucher codes.    |      |
| <b>Acceptance Criteria</b> | PASS: A valid point conversion produces a single-use voucher with a cryptographic checksum.<br>FAIL: An expired or duplicate voucher is redeemed. |      |
| <b>Rationale</b>           | Ensures valid loyalty-point conversion and secure gift-card redemption.                                                                           |      |

| FR-002                     | Functional                                                                                                                                                     | HIGH |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b>         | The system shall allow an account holder to aggregate loyalty points from supported merchant partners.                                                         |      |
| <b>Acceptance Criteria</b> | PASS: Points from a supported merchant are successfully added to the account holder's loyalty balance.<br>FAIL: Points from an unsupported merchant are added. |      |
| <b>Rationale</b>           | Enables users to combine loyalty points from multiple supported merchants.                                                                                     |      |

| FR-003                     | Functional                                                                                                                                                                                                         | HIGH |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b>         | The system shall calculate universal exchange credits using the current dynamic conversion rate for the user's loyalty points.                                                                                     |      |
| <b>Acceptance Criteria</b> | PASS: The system converts the user's loyalty points using the currently applicable conversion rate and displays the resulting exchange credits.<br>FAIL: The system uses an outdated or incorrect conversion rate. |      |
| <b>Rationale</b>           | Ensures loyalty points are converted accurately according to dynamic exchange rates.                                                                                                                               |      |

| FR-004                     | Functional                                                                                                                                                                      | HIGH |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b>         | The system shall allow an account holder to redeem eligible digital gift cards using universal exchange credits.                                                                |      |
| <b>Acceptance Criteria</b> | PASS: A gift card is issued when the account holder has sufficient exchange credits.<br>FAIL: Redemption is rejected when the account holder has insufficient exchange credits. |      |
| <b>Rationale</b>           | Provides the core gift-card redemption functionality of the platform.                                                                                                           |      |

| FR-005                     | Functional                                                                                                                                | HIGH |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b>         | The system shall lock a digital voucher during redemption and prevent its duplicate or expired redemption.                                |      |
| <b>Acceptance Criteria</b> | PASS: A voucher is locked during redemption and cannot be redeemed again.<br>FAIL: An expired or previously redeemed voucher is accepted. |      |
| <b>Rationale</b>           | Prevents fraudulent and duplicate gift-card redemption.                                                                                   |      |

## A.2 Non-Functional Requirements

| NFR-001            | Non-Functional                                                                                                                                             | HIGH |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b> | Point conversion and gift-card redemption operations must process within 250 ms with zero ledger balance drift.                                            |      |
| <b>Acceptance</b>  | Benchmarking tests confirm that point conversion and gift-card redemption meet the 250 ms target with zero ledger balance drift under simulated peak load. |      |

| NFR-002            | Non-Functional                                                                                                                                                | HIGH |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Requirement</b> | The system shall ensure that each generated digital voucher can be redeemed only once and shall reject expired or previously redeemed vouchers.               |      |
| <b>Acceptance</b>  | PASS: A valid voucher is accepted once and an expired or previously redeemed voucher is rejected.<br>FAIL: The system accepts a duplicate or expired voucher. |      |

## PART B

## Part B — Agile Backlog Interpretation

The Lab 1 functional requirements were grouped into three Epics and translated into five Jira User Stories. The User Stories below are Agile interpretations of the requirements, written in user-story format — they are not literal sentences copied from the Lab 1 requirements table.

### B.1 Epics

| #      | Epic Name                               | Purpose                                                                                                                   | Mapped Req.    |
|--------|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------|
| Epic 1 | Loyalty Point Management                | Manage aggregation of loyalty points from supported merchant partners.                                                    | FR-002         |
| Epic 2 | Point Conversion & Gift Card Generation | Handle validation/conversion of loyalty points into universal exchange credits/tokens and generation of digital vouchers. | FR-001, FR-003 |
| Epic 3 | Gift Card Redemption & Voucher Security | Handle eligible gift-card redemption and secure voucher locking and prevention of duplicate/expired redemption.           | FR-004, FR-005 |

### B.2 User Stories

| Story 1 · Jira Key: SE-4 |                                                                                                                                                    | Aggregate Loyalty Points |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Epic                     | Loyalty Point Management                                                                                                                           |                          |
| User Story               | As an account holder, I want to aggregate loyalty points from supported merchant partners, so that I can combine my loyalty points in one account. |                          |
| Mapped Requirement       | FR-002                                                                                                                                             |                          |
| Priority                 | High                                                                                                                                               |                          |

| Story 2 · Jira Key: SE-5 |                                                                                                                                                                                  | Validate Points and Generate Voucher |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Epic                     | Point Conversion & Gift Card Generation                                                                                                                                          |                                      |
| User Story               | As an account holder, I want the system to validate my loyalty points and convert them into universal exchange tokens, so that I can receive a valid single-use digital voucher. |                                      |
| Mapped Requirement       | FR-001                                                                                                                                                                           |                                      |
| Priority                 | High                                                                                                                                                                             |                                      |

| Story 3 · Jira Key: SE-6 |                                                                                                                                                                         | Calculate Universal Exchange Credits |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Epic                     | Point Conversion & Gift Card Generation                                                                                                                                 |                                      |
| User Story               | As an account holder, I want my loyalty points converted using the current dynamic conversion rate, so that I receive the correct number of universal exchange credits. |                                      |
| Mapped Requirement       | FR-003                                                                                                                                                                  |                                      |
| Priority                 | High                                                                                                                                                                    |                                      |

| Story 4 · Jira Key: SE-7 Redeem Digital Gift Card |                                                                                                                                                   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epic</b>                                       | Gift Card Redemption & Voucher Security                                                                                                           |
| <b>User Story</b>                                 | As an account holder, I want to redeem an eligible digital gift card using universal exchange credits, so that I can obtain the gift card I want. |
| <b>Mapped Requirement</b>                         | FR-004                                                                                                                                            |
| <b>Priority</b>                                   | <b>High</b>                                                                                                                                       |

| Story 5 · Jira Key: SE-8 Secure Voucher Redemption |                                                                                                                                                                                |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Epic</b>                                        | Gift Card Redemption & Voucher Security                                                                                                                                        |
| <b>User Story</b>                                  | As an account holder, I want the digital voucher to be locked during redemption and protected from duplicate or expired redemption, so that my gift-card redemption is secure. |
| <b>Mapped Requirement</b>                          | FR-005                                                                                                                                                                         |
| <b>Priority</b>                                    | <b>High</b>                                                                                                                                                                    |

### B.3 Prioritization

All five functional requirements are marked **HIGH** priority in the Lab 1 requirements table. Consequently, all five corresponding Jira User Stories (SE-4 through SE-8) were also set to **High** priority, since each maps directly to a High-priority requirement.

## PART C

## Part C — Story Point Estimates

The Lab 2 handout does not prescribe specific Story Point values for each User Story. It requires the use of the Fibonacci scale (0, 1, 2, 3, 5, 8, 13, 20, 40, 100) and states that Story Points should be based on **complexity, time, and uncertainty** — consistent with the Planning Poker concept. The values below are the Agile estimates chosen for this simulation and are not values mandated by the Lab 1 requirements document.

| Key  | User Story                           | Story Points | Estimation Rationale                                                                      |
|------|--------------------------------------|--------------|-------------------------------------------------------------------------------------------|
| SE-4 | Aggregate Loyalty Points             | 3            | Lower complexity — combining points from supported merchants.                             |
| SE-5 | Validate Points and Generate Voucher | 8            | Higher complexity — validation logic, token conversion, and secure voucher generation.    |
| SE-6 | Calculate Universal Exchange Credits | 5            | Moderate complexity — dynamic conversion-rate calculation.                                |
| SE-7 | Redeem Digital Gift Card             | 5            | Moderate complexity — eligibility and credit-sufficiency checks.                          |
| SE-8 | Secure Voucher Redemption            | 8            | Higher complexity — voucher locking and duplicate/expiry prevention (security-sensitive). |

**Total Estimated Backlog****29 Story Points**

These estimates represent the relative effort, complexity, and uncertainty of each story as agreed during the simulated Planning Poker exercise, using the Fibonacci scale specified in the Lab 2 handout. They are planning estimates for this exercise and are not derived from, or prescribed by, the Lab 1 requirements document.

## PART D

## Part D — Sprint Plan

The Lab 2 handout requires a Sprint 1 of 1-week duration, with selected backlog items progressed through the **To Do** → **In Progress** → **Done** workflow, and states that the exercise should conclude with 2 sprints. The sprint allocations below reflect the Agile sprint-planning decisions made for this simulation.

### D.1 Sprint 1

|               |                                                                     |
|---------------|---------------------------------------------------------------------|
| Duration      | 1 week                                                              |
| Sprint Goal   | Complete loyalty point aggregation and point conversion activities. |
| Planned Total | 16 Story Points                                                     |

| Key  | User Story                           | Story Points |
|------|--------------------------------------|--------------|
| SE-4 | Aggregate Loyalty Points             | 3            |
| SE-5 | Validate Points and Generate Voucher | 8            |
| SE-6 | Calculate Universal Exchange Credits | 5            |

### D.2 Sprint 2

|               |                 |
|---------------|-----------------|
| Planned Total | 13 Story Points |
|---------------|-----------------|

| Key  | User Story                | Story Points |
|------|---------------------------|--------------|
| SE-7 | Redeem Digital Gift Card  | 5            |
| SE-8 | Secure Voucher Redemption | 8            |

*The Sprint 1 and Sprint 2 allocations shown above are the selected Agile sprint plan for this simulation. The original Lab 1 requirements document does not prescribe which stories belong in which sprint.*

### D.3 Sprint Board Evidence

The screenshot below shows the Jira backlog with the created Epics and User Stories prior to sprint selection.



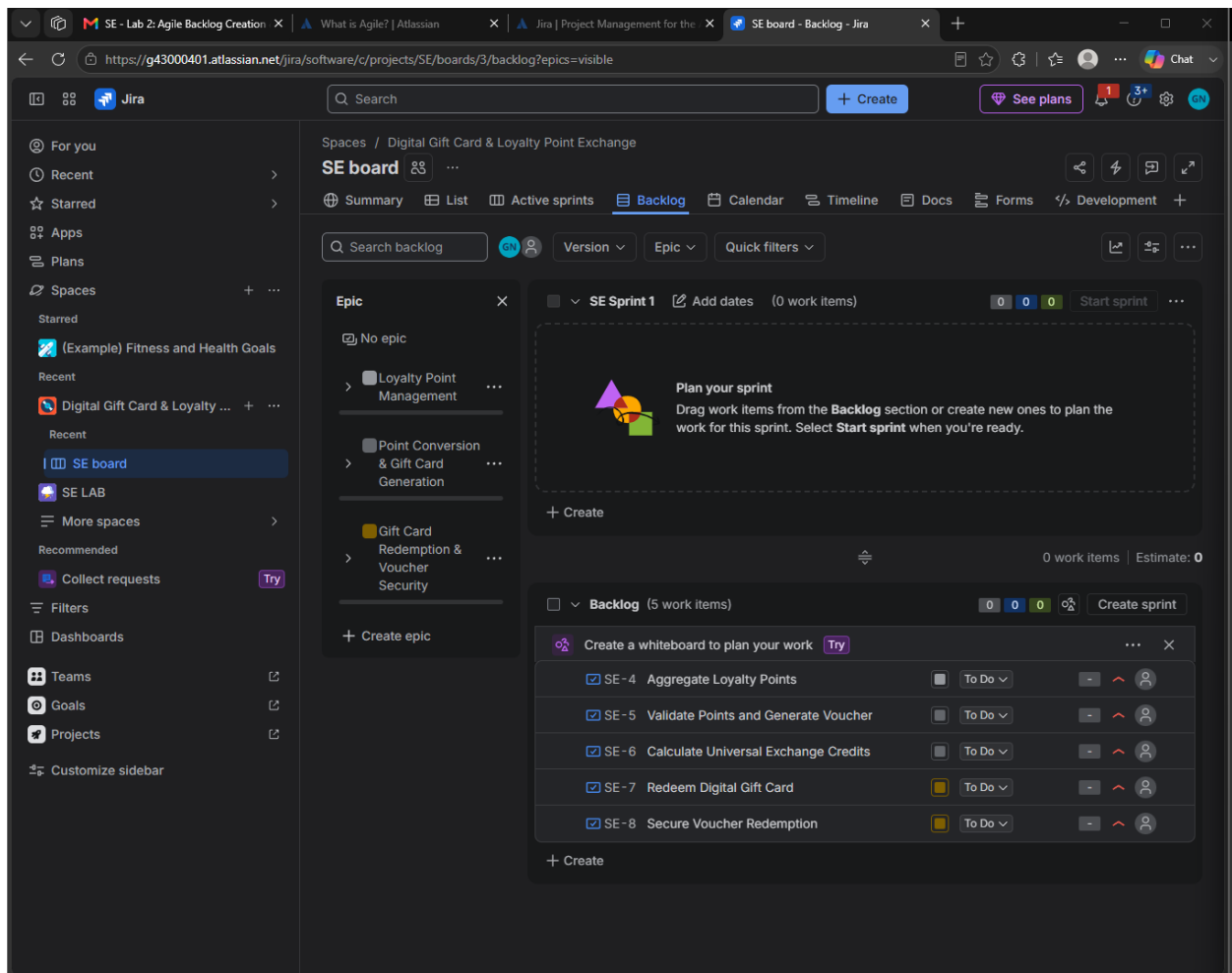


Figure D.1 — Jira Backlog view showing Epics (Loyalty Point Management; Point Conversion & Gift Card Generation; Gift Card Redemption & Voucher Security) and the five User Stories (SE-4 to SE-8).

The screenshot below shows the Active Sprint board for Sprint 1 with SE-4, SE-5 and SE-6 placed in the To Do column, ready to progress through To Do → In Progress → Done.

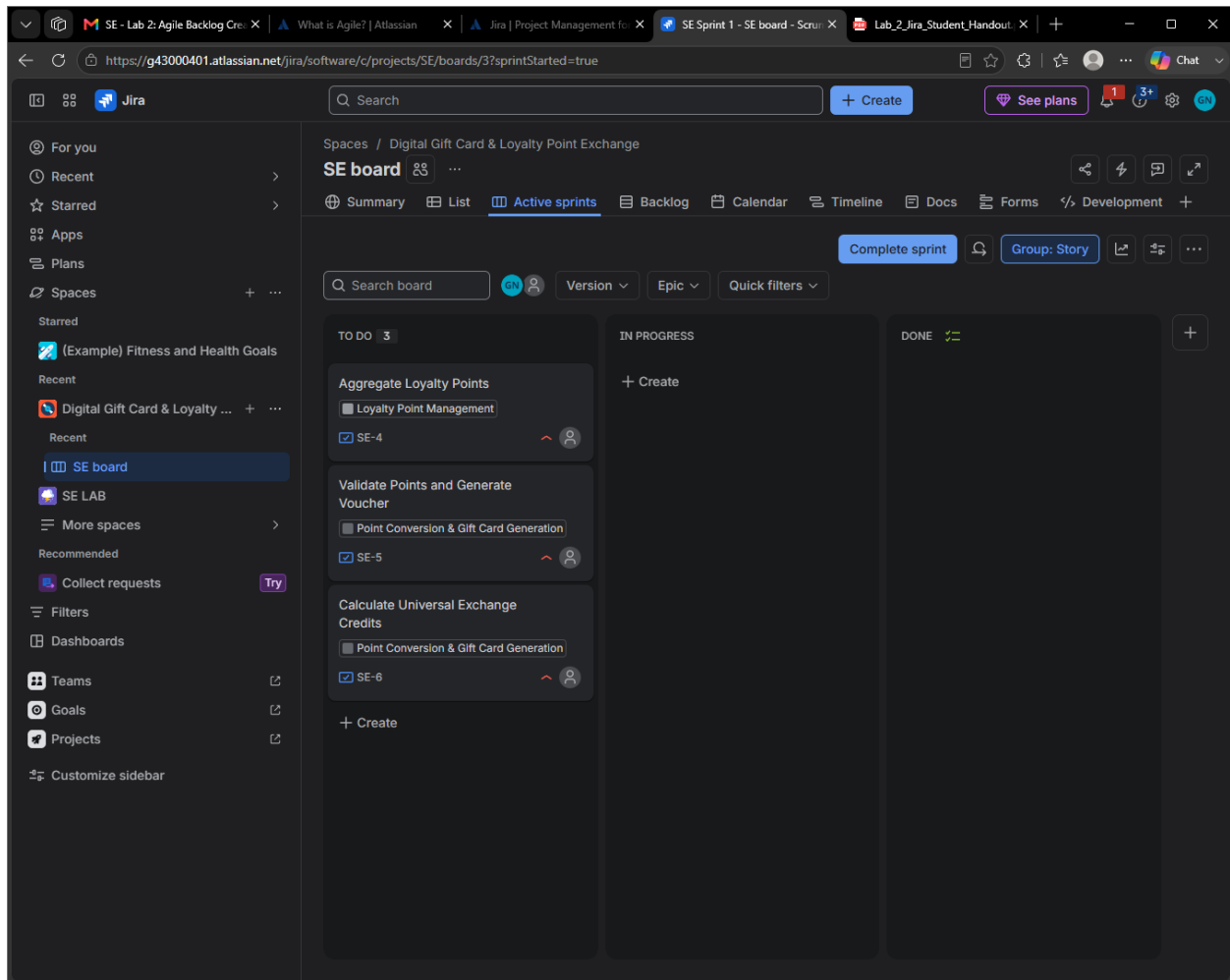


Figure D.2 — Jira Active Sprint board (Sprint 1) showing SE-4, SE-5 and SE-6 in the To Do column.

## PART E

## Part E — Burndown Chart & Sprint Execution

The Lab 2 handout requires observing the Jira **Reports** → **Burndown Chart** to see how work is completed across the sprint. A burndown chart visualizes work completed, total work remaining, and progress against the available sprint time.

**Chart basis:** The Jira Burndown Chart captured for this project is generated using **Issue Count** (as labelled on the chart's Y-axis and toolbar selector), not Story Points. This report presents it honestly as an Issue Count burndown; no Story Point burndown dataset was generated or is claimed here.

### E.1 Burndown Chart (Issue Count)

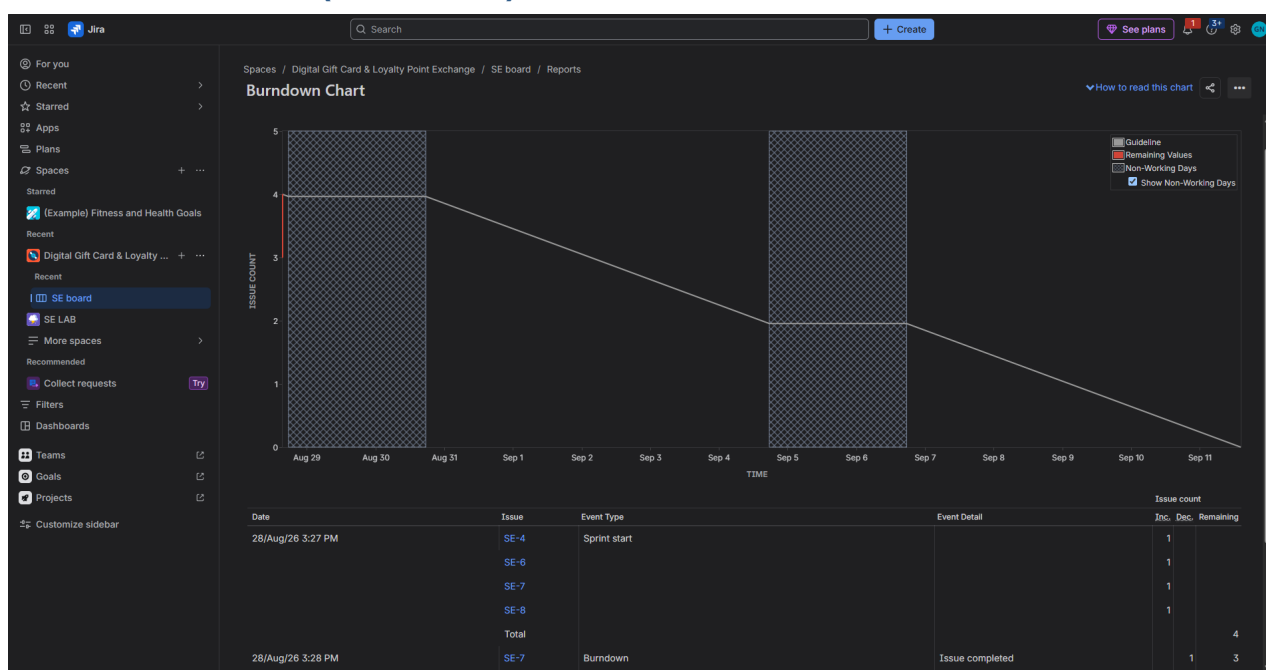


Figure E.1 — Jira Burndown Chart (Issue Count) for the active sprint, showing the issue-count guideline, remaining values, and the sprint event log (sprint start, burndown of SE-7, and sprint end).

The event log accompanying the chart records that the sprint started with 4 issues (SE-4, SE-6, SE-7, SE-8), that SE-7 was subsequently marked as completed (burndown event), and that the sprint was later ended by Girish G N with SE-4 and SE-6 carried over and SE-7 completed.

### E.2 Sprint Progression Evidence

The backlog view below reflects the state after Sprint 1 was completed and a second sprint window (Sprint 2, 4 Sep – 11 Sep) was created, carrying forward SE-4 (In Progress) and SE-6 (In Review), with SE-7 and SE-8 in the Backlog before being added to Sprint 2.

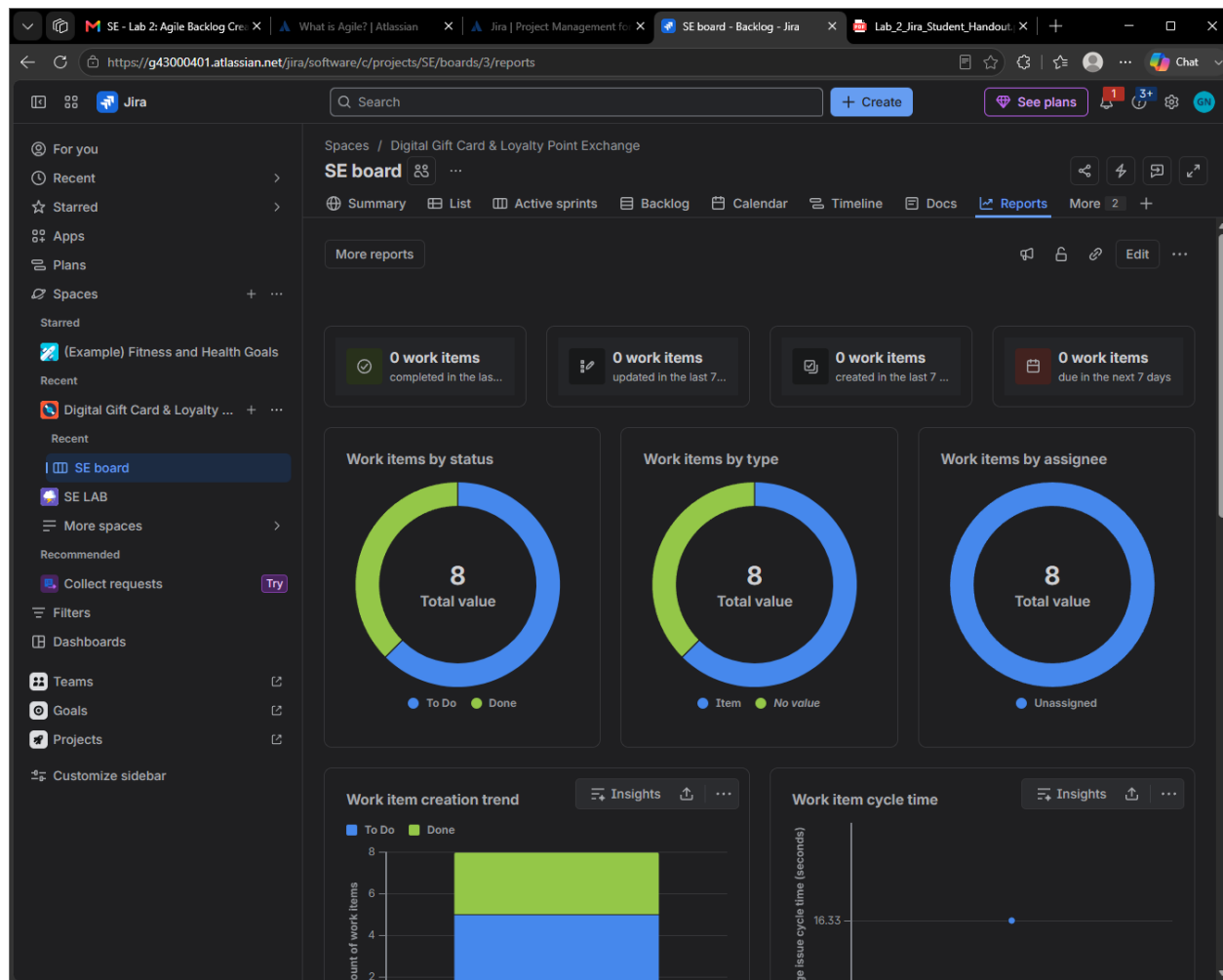


Figure E.2 — Jira Reports overview showing Work Items by Status, by Type, and by Assignee for the SE board (8 total work items).

**PART F**

## Part F — Reflection

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### 1. Did your estimations reflect the actual effort?

The Fibonacci estimates provided a useful relative measure of complexity and uncertainty. Some items were larger because they involved validation, voucher generation, dynamic conversion, or security handling. Since this was a simulated sprint, the estimates are best interpreted as planning estimates rather than measurements of real development effort.

### 2. Was your backlog well-prioritized?

Yes. The functional requirements were all marked High priority in Lab 1 because they represent the core functionality of the digital gift-card and loyalty-point exchange system. The backlog was therefore prioritized according to business and user impact.

### 3. How did your simulated sprint align with your plan?

The sprint simulation followed the planned Agile workflow by selecting backlog items, placing them into the sprint, and progressing them through To Do, In Progress, and Done. The sprint was organized around the major functional areas identified from the Lab 1 requirements.

### 4. What insights did the burndown chart give about your team's capacity?

The burndown chart provides a visual indication of how quickly work is being completed relative to the available sprint time. A steady decline indicates consistent progress, while a sharp drop may indicate that several items were completed together or that work was not broken down evenly. The chart can therefore help evaluate planning, workload, and team capacity.

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**Deliverables checklist (per Lab 2 handout):** (1) Jira backlog with Epics and User Stories — see Figure D.1; (2) Story Point assignments — see Part C; (3) Sprint board / Active Sprint view — see Figure D.2; (4) Burndown Chart — see Figure E.1; (5) this document, answering the reflection questions above.